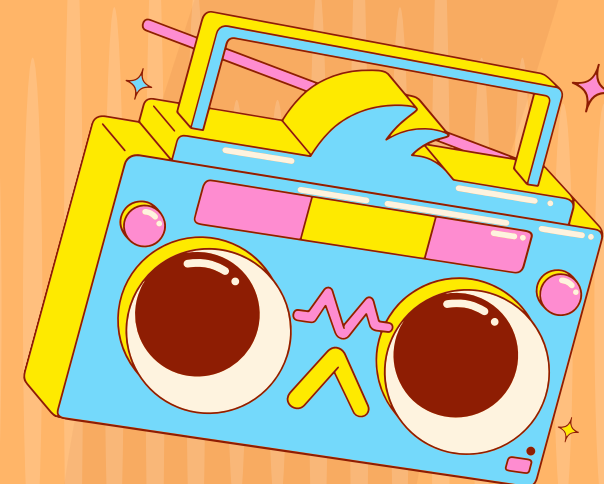


SPOTIFY

Data





Members

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- Natnicha Sukchuenanant 6580812
- Chanon Pluemhathaikij. 6581103
- Supakorn Panyadee 6581117



How we import data to hadoop?

```
ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-01-c28e7fbec31b/zones/us-east1-c/instances/ch6cluster-m?authuser=0&hl=en_US&projectNumber=464272641239&useAdminProxy=true&pageViewId=C39A3541-4AB9-44DD-A33B-A29EC2C53BC9 - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-01-c28e7fbec31b/zones/us-east1-c/instances/ch6cluster-m?authuser=0&hl=en_US&projectNumber=464272641239&useAdminProxy=true&pageViewId=C39A3541-4AB9-44DD-A33B

SSH-in-browser

IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=27292931
File Output Format Counters
  Bytes Written=5140
2025-06-05 17:24:15,286 INFO streaming.StreamJob: Output directory: gs://mynewbucket111/outputPop/
student-03-6e7c8329e36a@ch6cluster-m:~$ hadoop jar /usr/lib/hadoop/hadoop-streaming-3.3.6.jar \
-files \
  gs://mynewbucket111/input/mapperYear.py,\
gs://mynewbucket111/input/YearReducer.py \
-mapper "python3 mapperYear.py" \
-numReduceTasks 1 \
-reducer "python3 YearReducer.py" \
-input gs://mynewbucket111/input/spotify-data.csv \
-output gs://mynewbucket111/outputYear/
2025-06-05 17:34:59,844 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-06-05 17:34:59,948 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-06-05 17:34:59,948 INFO impl.MetricsSystemImpl: google-hadoop-file-system metrics system started
packageJobJar: [] [/usr/lib/hadoop/hadoop-streaming-3.3.6.jar] /tmp/streamjob6721667377345894340.jar tmpDir=null
2025-06-05 17:35:01,550 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at ch6cluster-m.us-east1-c.c.qwiklabs-gcp-01-c28e7fbec31b.internal./10.142.0.4:8032
2025-06-05 17:35:01,810 INFO client.AHSProxy: Connecting to Application History server at ch6cluster-m.us-east1-c.c.qwiklabs-gcp-01-c28e7fbec31b.internal./10.142.0.4:10200
2025-06-05 17:35:02,395 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at ch6cluster-m.us-east1-c.c.qwiklabs-gcp-01-c28e7fbec31b.internal./10.142.0.4:8032
2025-06-05 17:35:02,396 INFO client.AHSProxy: Connecting to Application History server at ch6cluster-m.us-east1-c.c.qwiklabs-gcp-01-c28e7fbec31b.internal./10.142.0.4:10200
2025-06-05 17:35:02,726 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/student-03-6e7c8329e36a/.staging/job_1749141604797_0006
2025-06-05 17:35:03,523 INFO mapred.FileInputFormat: Total input files to process : 1
2025-06-05 17:35:03,595 INFO mapreduce.JobSubmitter: number of splits:21
2025-06-05 17:35:03,825 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1749141604797_0006
2025-06-05 17:35:03,825 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-06-05 17:35:04,056 INFO conf.Configuration: resource-types.xml not found
2025-06-05 17:35:04,057 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-06-05 17:35:04,148 INFO impl.YarnClientImpl: Submitted application application_1749141604797_0006
2025-06-05 17:35:04,198 INFO mapreduce.Job: The url to track the job: http://ch6cluster-m.us-east1-c.c.qwiklabs-gcp-01-c28e7fbec31b.internal.:8088/proxy/application_1749141604797_0006/
2025-06-05 17:35:04,200 INFO mapreduce.Job: Running job: job_1749141604797_0006
2025-06-05 17:35:14,373 INFO mapreduce.Job: Job job_1749141604797_0006 running in uber mode : false
2025-06-05 17:35:14,375 INFO mapreduce.Job:  map 0% reduce 0%
2025-06-05 17:35:32,555 INFO mapreduce.Job:  map 14% reduce 0%
2025-06-05 17:35:36,584 INFO mapreduce.Job:  map 24% reduce 0%
2025-06-05 17:35:37,590 INFO mapreduce.Job:  map 33% reduce 0%
2025-06-05 17:35:47,655 INFO mapreduce.Job:  map 43% reduce 0%
2025-06-05 17:35:48,662 INFO mapreduce.Job:  map 48% reduce 0%
2025-06-05 17:35:57,716 INFO mapreduce.Job:  map 67% reduce 0%
2025-06-05 17:36:03,762 INFO mapreduce.Job:  map 81% reduce 0%
2025-06-05 17:36:14,826 INFO mapreduce.Job:  map 86% reduce 0%
2025-06-05 17:36:17,844 INFO mapreduce.Job:  map 100% reduce 0%
2025-06-05 17:36:27,900 INFO mapreduce.Job:  map 100% reduce 100%
2025-06-05 17:36:29,922 INFO mapreduce.Job: Job job_1749141604797_0006 completed successfully
2025-06-05 17:36:30,024 INFO mapreduce.Job: Counters: 60
File System Counters
  FILE: Number of bytes read=2038674
  FILE: Number of bytes written=10480810
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  GS: Number of bytes read=27292931
  GS: Number of bytes written=5218
  GS: Number of read operations=6677
  GS: Number of large read operations=0
  GS: Number of write operations=404
  HDFS: Number of bytes read=1974
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=21
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=0
```



Mapper

reducer.py

mapper.py

D: > oscar > mapper.py > ...

```
1  #!/usr/bin/env python3
2  import csv
3  import sys
4
5  reader = csv.reader(sys.stdin)
6  for index, row in enumerate(reader):
7      if(index != 0):
8          print(f"{row[2]}\n") #since in our .csv file the artists column is index 2
```

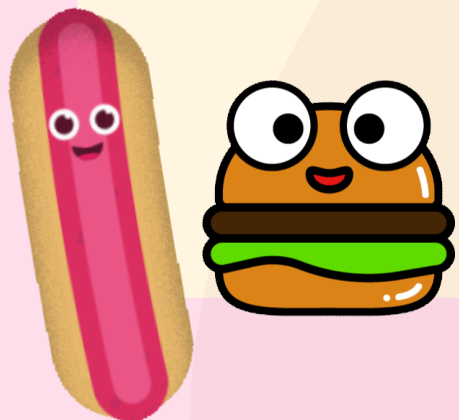
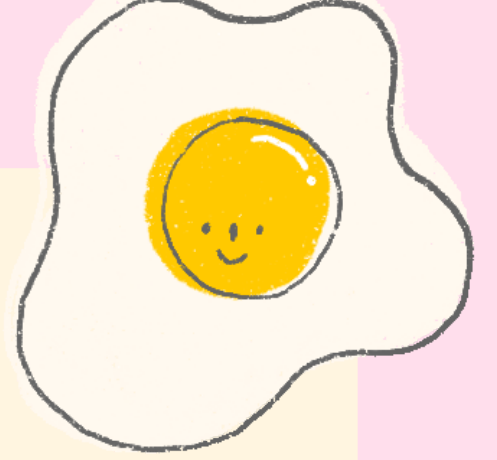


Reducer

```
reducer.py X mapper.py ●
D: > oscar > reducer.py > ...
1  #!/usr/bin/env python3
2  import sys
3  import re
4  from collections import defaultdict
5
6  counts = defaultdict(int)
7
8  #with open("/content/drive/MyDrive/Big Data Project(Spotify)/demofile.txt", mode='r') as file:
9
10 for line in sys.stdin:
11
12     word = line.strip()
13
14     cleanWord = word.replace("\\", "")
15
16     try:
17         mylist = eval(cleanWord)
18     except Exception:
19         mylist = [cleanWord]
20
21     #Check some unstructure format
22     final_artists = []
23     for entry in mylist:
24         pieces = re.split(r'[/,-]', entry.strip())
25         for p in pieces:
26             cleaned = p.strip()
27             if cleaned:
28                 final_artists.append(cleaned)
29
30     #Separate all the artist
31     for artist in final_artists:
32         counts[artist] += 1
33
34 for artist, count in counts.items():
35     print(f"{artist:60}{counts[artist]}")
```

Output

'In The Heights' Original Broadway Company	9
Mandy Gonzalez	9
'Legally Blonde' Ensemble	9
Michael Rupert	2
Nikki Snelson	1
'Til Tuesday	8
Adolescent's Orquesta	9
Alexander 'Skip' Spence	7
Alexander O'Neal	14
Alexis Korner's Blues Incorporated	1
Alive 'N Kickin'	1
Amy Hanaiali'i	1
Willie K	4
Anita O'Day	103
Billy May Orchestra	1
Gene Krupa & His Orchestra	5
Roy Eldridge	20
Marty Paich & His Orchestra	6
Oscar Peterson Quartet	15
Russ Garcia and His Orchestra	3
The Gary McFarland Orchestra	2
The Three Sounds	10
Auli'i Cravalho	5
Vai Mahina	2
Olivia Foa'i	3
Opetia Foa'i	4
Matthew Ineleo	2
B. D'Addario as Flounder	1
Mersisters	1
The Little Mermaid Original Broadway Cast	1
Bailey's Nervous Kats	8
Ballin' Jack	1





Why do we choose the Artist's column



Database

CHANON'S ORG - 2025-05-05 > PROJECT 0 > DATABASES

 SpotifyCluster

VERSION
8.0.9

REGION
GCP Singapore (asia-southeast1)

Overview

Real Time

Metrics

Collections

Atlas Search

Query Insights

Performance Advisor

Online Archive

DATABASES: 2 COLLECTIONS: 7

PREVIEW

New Data Explorer

 VISUALIZE YOUR DATA

 REFRESH

+ Create Database

 Search Namespaces

▼ ArtistInfo

| Artists

▶ sample_mflix

ArtistInfo.Artists

STORAGE SIZE: 1.1MB LOGICAL DATA SIZE: 1.56MB TOTAL DOCUMENTS: 27984 INDEXES TOTAL SIZE: 828KB

Find

Indexes

Schema Anti-Patterns 0

Aggregation

Search Indexes

[Generate queries from natural language in Compass](#)

INSERT DOCUMENT

Filter 

Type a query: { field: 'value' }

Reset

Apply

Options ▶

QUERY RESULTS: 1-20 OF MANY

```
{
  "_id": ObjectId('6845589ef42928cfbedcf22e'),
  "name": "'In The Heights' Original Broadway Company",
  "amount": 9
}
```

```
{
  "_id": ObjectId('6845589ef42928cfbedcf22f'),
  "name": "Mandy Gonzalez",
  "amount": 9
}
```



Database

```
[ ] SampleDb = client.ArtistInfo
    user = SampleDb.Artists
```



```
[ ] #Upload our Hadoop output into the MongoDB via Google Colab
    with open("/content/drive/MyDrive/BigData/Artist_output_part-00000") as f:
        for line in f:
            cols = line.split()
            length = len(cols)
            name = ""
            num = 0

            for i in range(length):
                if (i == length-1):
                    num += int(cols[i])
                else:
                    name += " "+cols[i]

            #insert each artist
            user.insert_one({
                'name' : name.strip(),
                'amount' : num
            })
```



Top 10 artists with the most songs

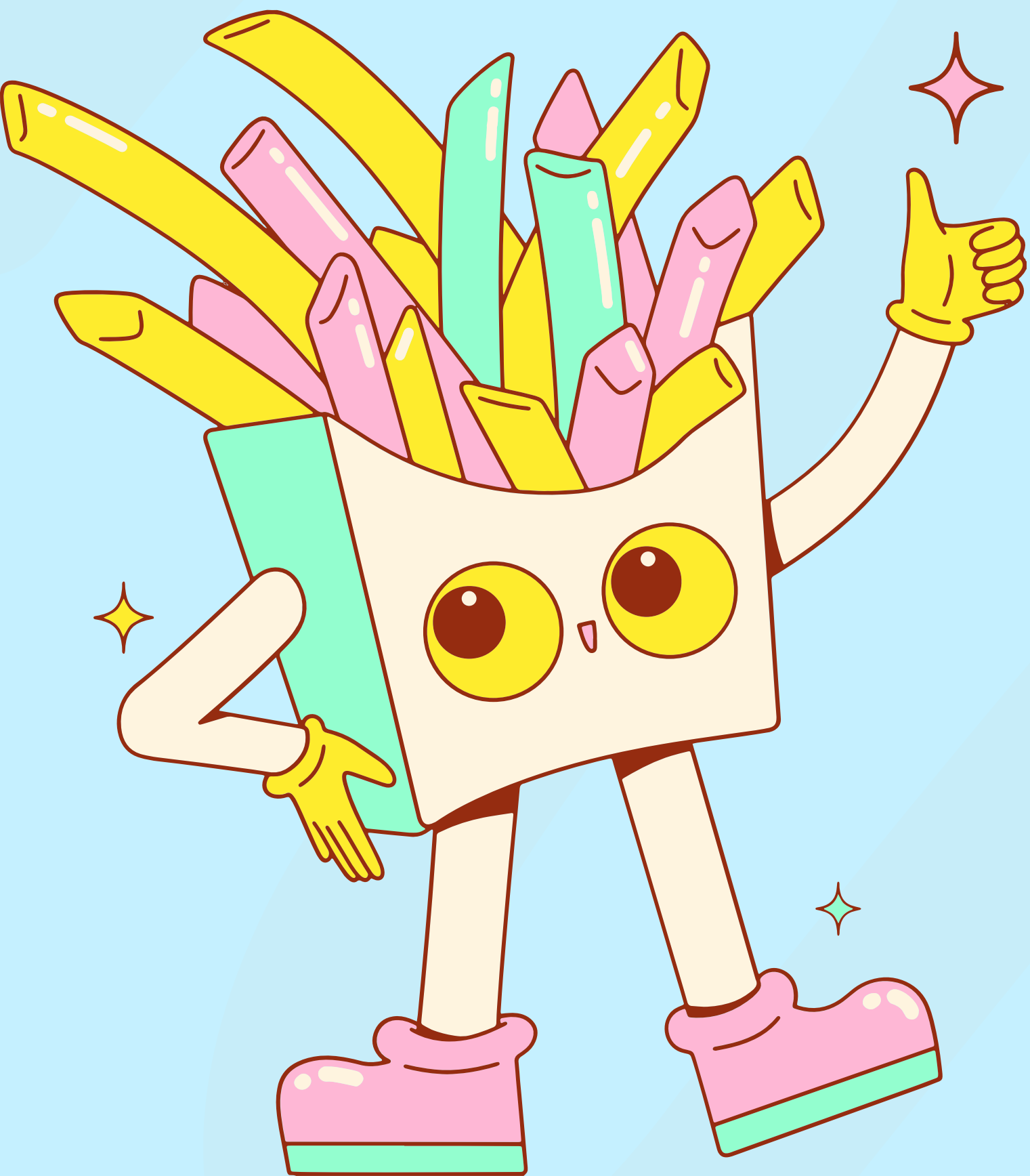
```
✓ 0s ▶ Top10 = list(user.aggregate([
    {
        "$sort":{"amount":-1}
    },
    {
        "$limit":10
    }
]))

for e in Top10:
    print(e)
```

♥ ♥ ♥ ♥ ♥
GAME OVER

```
⇒ {'_id': ObjectId('68456142f42928cfbedd1985'), 'name': 'Francisco Canaro', 'amount': 2234}
{'_id': ObjectId('6845708cf42928cfbedd5f30'), 'name': 'Эрнест Хемингуэй', 'amount': 1214}
{'_id': ObjectId('68455a6af42928cfbedcfa64'), 'name': 'Frédéric Chopin', 'amount': 1035}
{'_id': ObjectId('68456753f42928cfbedd3500'), 'name': 'Ludwig van Beethoven', 'amount': 975}
{'_id': ObjectId('68456fb0f42928cfbedd5b3f'), 'name': 'Wolfgang Amadeus Mozart', 'amount': 937}
{'_id': ObjectId('68455a6cf42928cfbedcfa6c'), 'name': 'Johann Sebastian Bach', 'amount': 868}
{'_id': ObjectId('6845708cf42928cfbedd5f2f'), 'name': 'Эрих Мария Ремарк', 'amount': 781}
{'_id': ObjectId('684563acf42928cfbedd246b'), 'name': 'Igor Stravinsky', 'amount': 739}
{'_id': ObjectId('684559d5f42928cfbedcf7bb'), 'name': 'Frank Sinatra', 'amount': 732}
{'_id': ObjectId('68455a29f42928cfbedcf93a'), 'name': 'Giuseppe Verdi', 'amount': 691}
```





THANK YOU

